

Review

Fresh Insight through a Keynesian Theory Approach to Investigate the Economic Impact of the COVID-19 Pandemic in Pakistan

Kashif Abbass ^{1,2} , Halima Begum ^{3,*} , A. S. A. Ferdous Alam ⁴ , Abd Hair Awang ⁵ ,
Mohammed Khalifa Abdelsalam ⁶ , Ibrahim Mohammed Massoud Egdair ⁷  and Ratnaria Wahid ⁴

- ¹ School of Economics and Management, Nanjing University of Science and Technology, Nanjing 210094, China; kashifabbass@njjust.edu.cn
 - ² Riphah School of Business and Management, Riphah International University, Lahore 54000, Pakistan
 - ³ Schools of Economics, Finance, and Banking, Universiti Utara Malaysia, Sintok 06010, Kedah, Malaysia
 - ⁴ School of International Studies, Universiti Utara Malaysia, Sintok 06010, Kedah, Malaysia; ferdous@uum.edu.my (A.S.A.F.A.); ratnaria@uum.edu.my (R.W.)
 - ⁵ Faculty of Social Sciences & Humanities, Universiti Kebangsaan Malaysia, Bangi 43600, Selangor, Malaysia; hair@ukm.edu.my
 - ⁶ Department of Banking and Risk Management, School of Economics, Finance and Banking, Universiti Utara Malaysia, Sintok 06010, Kedah, Malaysia; khalifaedri@gmail.com
 - ⁷ Department of Business Administration, Faculty of Economic and Accounting-Murzq, Sebha University, Sebha 00218, Libya; Ibr.egdair@sebhau.edu.ly
- * Correspondence: dr.halima.begum@uum.edu.my; Tel.: +60-182-487-410



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Abstract: Beyond the immediate impositions of dealing with COVID-19, this disease represents a severe and significant challenge confronting Pakistan's economy. The study's objective was to evaluate the coronavirus epidemic's effect on Pakistan's economy and measures devised to mitigate the damage done by this disease. The study research design used the elementary concept of Keynesian theory comprising of the mapping of systematic behavior of the COVID-19 pandemic. Issues were formally underpinned, described, and visualized through the Keynesian theory concept. The eruption of COVID-19 has jolted the national and international economy. Pakistan is included, causing millions of people to stay at home, lose their jobs, and suspend or end business operations. Unemployment in Pakistan has reached nearly 25 million people, driving many towards conditions of hunger and poverty as the major economic damage in several sectors is anticipated at around PKR 1.3 trillion. The hardest-affected sectors comprise industries such as tourism and travel, financial markets, entertainment, manufacturing, etc., having a devastating effect on gross domestic product (GDP). It is mainly daily-wage earners and people running small businesses that have been seriously exploited and subjected to a curfew-like situation. However, the Keynesian theory suggests that supportive macroeconomic policies must restore trust, demand recovery, and provide interest-free loans to overcome Pakistan's currently upcoming crisis.

Keywords: COVID-19; Pakistan economy; economic scale; Keynesian theory; mitigation; economic revival measures

1. Introduction

In recent decades, normal economic activities have become threatened and radicalized by expanding human enterprises that are risky and imprudent [1–4]. Consequently, the natural environment is at significant risk and impacts people's communities [5,6]. Historically, the dangers to human wellbeing originate in people's contact with the natural habitat and affect it in such a way as to increase the risk and spread of infections that are difficult to control [7]. Thus, on a global scale, the world has, at times, not resisted certain diseases and sicknesses, known and obscure, despite the tremendous medical advances made in

eliminating lethal organisms in the 1970s [8]. For instance, the World Bank group in China (2020) [9] only recently described China as one of the world's cloudiest nations due to its vast population and despite now being very wealthy [10]. The authors [11–13] reported that China has been susceptible to spreading diseases like African pig fever, which is highly transmissible in the form of influenza, rabies, plague, and other zoonotic pathogens that are potentially pandemic [14–16]. The authors [17] stated that the most probable cause of the COVID-19 infection was the bats, which ended up being sold at a Wuhan market, where interactions between humans and live creatures occurred. This subsequent infection spread from China's Hubei area and affected around 211,000 individuals by 31 December 2019 [18], and it rapidly spread from human to human on a global scale. Previously known as the "2019 Novel Coronavirus," it became the infamous COVID-19 [19,20], though vaccines have been invented to treat victims.

Nonetheless, the world's economies have been now extensively damaged since March 2020 due to this pandemic, and the response needs to avoid a possible health catastrophe [21,22]. Several studies document evidence that the economic disadvantages or destruction associated with this epidemic are worldwide [23–28]. The epidemic's effects go beyond death and indisposition and have an enormous impact on economic systems with dangerous social consequences [26]. More than 100 countries are now affected, with a fatality rate of 5.5% reported in recent weeks due to COVID-19 [29]. Due to its highly infectious nature, governments worldwide have implemented lockdowns in various measures, which means locking down economic activities [30]. As a result, [31] found that commercial, industrial, and professional activities are operating at less than half their previous rates due to restrictions on people's movements imposed by federal, provincial, regional, state, and local governments, which have the most impact on aggregate demand (AD) and supplies.

The Worldometer [18] evaluated that COVID-19 spread very quickly, with 248,983,180 cases, 5,040,413 deaths, and 225,552,585 recovered to date. It started in Wuhan, China, on 17 November 2019, and three weeks later, cases were reported in Japan, South Korea, and the U.S., which declared their first instances of COVID-19. It took 42 days for the confirmed cases to reach 100 in the U.S., while Japan and South Korea took 31 and 29 days, respectively. Elsewhere, India had 34,320,142 confirmed cases as of 8 November 2021, while Italy had over 4,782,802 [18]. Regarding COVID-19 issues, the U.K., the fourth most-devastated European nation, has outpaced France. Spain has more than 5,019,255 affirmed cases and is currently the second most infected E.U. nation. The number of cases in Germany, Russia, Turkey, and Brazil is 4,662,011, 8,673,860, 8,121,226, and 21,835,785, respectively [18]. Meanwhile, [32,33] reported that the infection's spread throughout the populace in different nations occurs at different rates, much depending on the nature of social, economic, and political variables, such as people's responses, the density of the population, the size of family units, etc.

Figure 1 below illustrates that the COVID-19 pandemic entered Pakistan on 26 February 2020, when an undergraduate returned from Iran to Karachi. By 18 March, each of the four regions, (the two autonomous regions), and Islamabad's government region had reported coronavirus cases. Meanwhile, more than 1,277,160 patients were confirmed by 8 November 2021, in Pakistan. However, Punjab had the most victims, with 441,176 reported, while Khyber Pakhtunkhwa suffered 178,643 casualties as of 8 November 2021. Nonetheless, after the partial lockdown, according to the government of Pakistan's official portal, there are a total number of cases of 1,277,160 and 28,547 deaths. It is obvious the cases and death rates are increasing rapidly. As a result, in order to save the country's economy, new strategies and policies must be implemented.

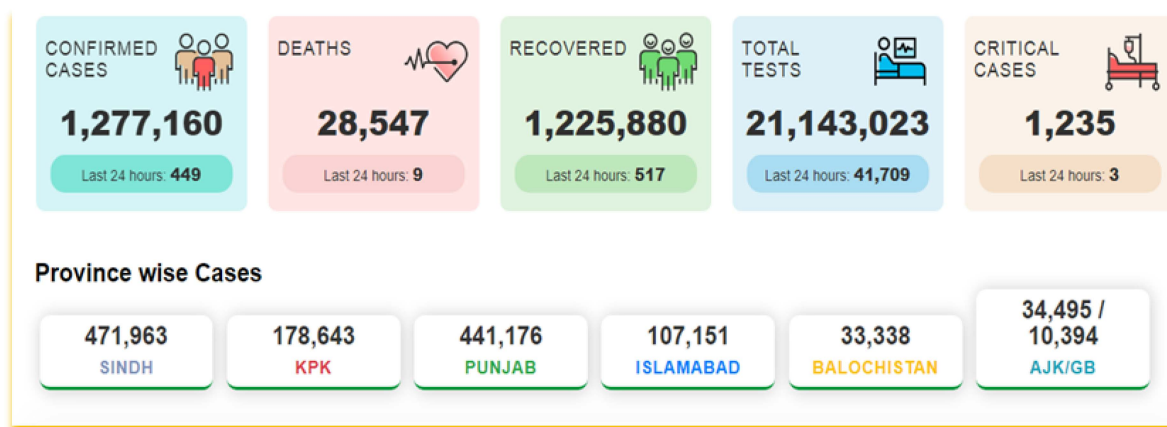


Figure 1. Cases of coronavirus in Pakistan. Source: [34].

It is important to note that contradictions have been observed among the governments, private-sector companies, and the public regarding the emergence of the pandemic above worldwide. However, it is pertinent that in many sections of Pakistan, particularly the provinces, a fierce debate has erupted concerning the impact of the coronavirus epidemic on the nation's economy [35,36]. Pakistan is experiencing challenges due to a lack of resources to address the current scenario. While the coronavirus (COVID-19) is being handled through systematic initiatives and policies, these are constrained by limited resources [37]. Besides, economic uncertainty could be lessened by following other countries' strategies, such as the authors [21] asserted in a context of uncertainty about the end of the pandemic and complemented by economic, fiscal, and monetary policies that mitigate the economic recession. The authors propose shifting strategies and initiatives from an indiscriminate suppression strategy to a targeted, effective, and intelligent mitigation strategy that minimizes the risk of human-life costs and socioeconomic costs.

The pandemic affects the world economy and creates a hazard to economic development among low- and middle-income countries like Pakistan [38,39]. Apart from the significant inconvenience of combating COVID-19, Pakistan, like all other countries, is struggling to find the funds to deal with this situation [40,41]. The authors opined that Pakistan's economy could overcome the vast challenges, while others author [42] suggested to come out with more advanced economic strategies and several theoretical and logical arguments to bolster aggregate demand.

The study outcome would suggest reliable opportunities for Pakistan and highlight the importance of international cooperation to promote cooperative and supportive macroeconomic policies and the benefits of joint macroeconomic actions. The critical aspect to be developed is "a best action" plan that helps Pakistan adopt the best strategy for post-COVID to promote better recovery development and related bilateral initiatives.

This study attempted to compute the impending Pakistani financial cost of COVID-19 under various probable situations. The objective was to offer direction to the experts and policymakers for internationally coordinated strategy retorts to tame the case. The study constructs the understanding obtained from gauging SARS's economics and the influenza epidemic's economics [43]. The research initially sheds light on Pakistan's different significant sectors most affected by COVID-19. Section 3 used Keynesian theory for theoretical support, stressing its strengths to measure the epidemic's international cost. In Section 4, the study development framework, like the type of policy challenges the Pakistani government faces and the kind of policy responses the Pakistani government took to mitigate the pandemic's effect, is described. Section 5 discusses the study's recommendations in several directions. Section 6 of the article determines the main findings of the study.

2. The Macroeconomic Effects of a Pandemic in Pakistan

Around the world, Pakistan's economy is struggling in a severe downturn. COVID-19's impacts on Pakistan's economy could result in a sharp drop in GDP growth [42], a skewed current and fiscal balance, disrupted supply chains, and increased unemployment across the board [37]. Several macroeconomic effects are explained in more detail below in Figure 2, considering these issues.

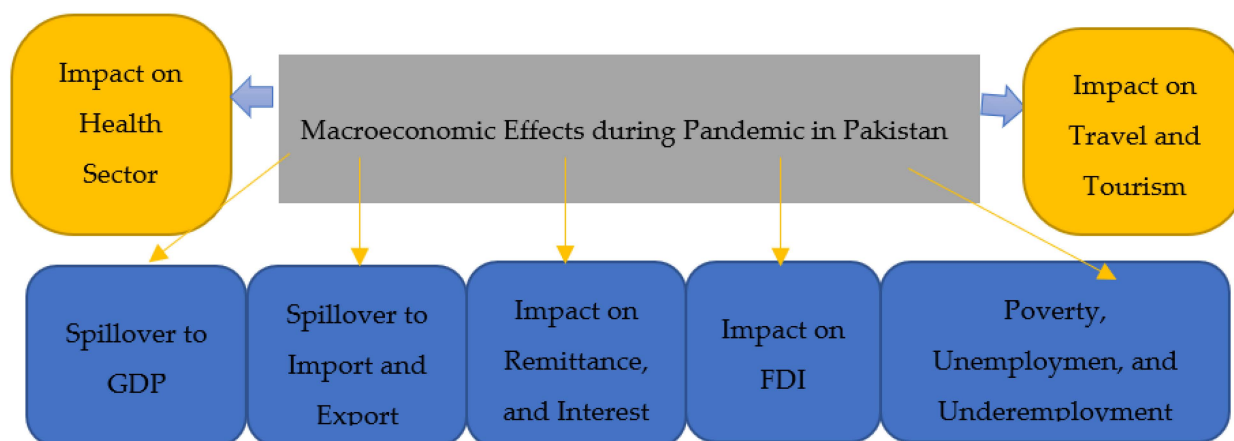


Figure 2. Several macroeconomic factors that impact Pakistan's economy. Source: author's development.

2.1. Spillover to Gross Domestic Product (GDP)

The worldwide pandemic has resulted in national output falling by 2% below the benchmark for the world. It is estimated that the manufacturing and technical sectors contribute 2.5% and 1.8%, respectively, to the national GDP. Surprisingly, local administrations were affected by the pandemic [10]. The global epidemic is anticipated to decrease China's GDP by 3.7% and Pakistan's GDP by 4.64% [38]. Subsequently, China's falling GDP has had "knock-on" effects on Pakistan's total aggregates because both countries exchange in trade [44]. The authors [45] reports that a fall in GDP leads to a subsequent decline in business (and capital), prompting lower manufacturing and productivity, lower import–export, and an across-the-board reduction in families' livelihoods, employment, and take-home pay. The total national output in Pakistan was worth USD 320 billion in 2019 [46], and the GDP was estimated to be 0.26% of the world [47]. In Figure 3, the bar chart estimates that GDP was expected to cause a 10% economic loss (estimated at PKR 1.1 tri.) during the last quarter (April–June 2020) of the 2020 financial year. As calculated by PIDE, if imports and exports fell by 20% during the epidemic, a 6.64% loss was observed in Pakistan's GDP [48,49].

2.2. Spillover to Import and Export of Pakistan

At present, Pakistan's imports have a 2:1 tariff restriction ratio [47], with oil accounting for 25% of the country's total import bill [50]. Recurring companies for individuals dependent on hardware imports, metals such as iron and steel, and other substances imported will almost certainly fail because no one wants to conduct business in the current environment [51]. Imports decreased by 16.9%, from USD 43,447 million in July–April 2018–2019 to USD 36,091 million in July–April 2019–2020 [52]. Pakistan is likely to see a decline in imports, which would have an impact on its GDP due to a lack of available materials.

Exports are also an essential aspect of a country's economy. World trade has dropped by 4.6%. Various economies that now have greater than mediocre global losses of business to the European region include China (9.8%), Hong Kong SAR, Singapore (8.5%), Cambodia (7.4%), the Russian Federation (7.3%), Lao PDR (7.3%), Thailand (6.8%), and the Philippines (6.4%). On the contrary, Europe, Canada, and the United States have been projected to decrease by 4.5% [10,53]. Pakistan's exports of textiles amount to 55% compared to fish, rice,

wheat, sports, and mineral fuels [47]. Exports are undeniably harmed when a pandemic occurs [51], and a 2.4% decrease means a loss of USD 19,650 million from July to April 2019–2020, compared to the previous fiscal year [52].

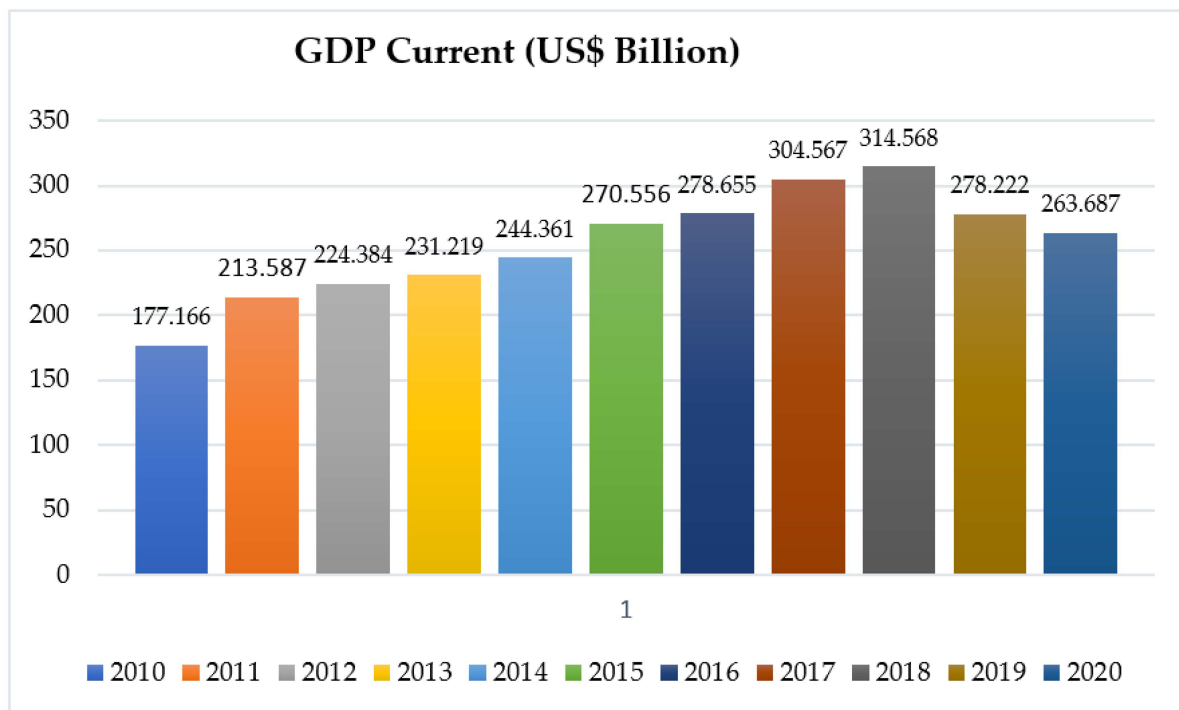


Figure 3. GDP of Pakistan. Source: [49].

2.3. The Effect on Remittances and Interest Rates

Remittances represented over 5% of GDP in 2019, and this applied to 66 nations, especially the developing economies [54–57]. As indicated by the State Bank of Pakistan [58], Pakistan’s internal settlements were down 4.4% m-o-m (monthly) by February 2020, at USD 1824 million during the COVID-19 pandemic. During 2020, most remittances sent to Pakistan came from the UAE (USD 388.1 million), KSA (USD 422.0 million), and the USA (USD 333.5 million). Undeniably, these kinds of payments have been a significant boost to Pakistan’s economy for many years, and, overall, USD 1 billion has been sent every month over the last two decades, cresting in the 2018 financial year when settlements were worth approximately USD 20 billion [59,60]. The World Bank group [61] noted that sending remittances may be mainly due to the widespread shutdowns caused by COVID-19 outbreaks and that these financial settlements to Pakistan will decline [48]. Moreover, the SBP’s Monetary Policy Committee (MPC) amended the benchmark interest rate in March 2020 as financing costs rose by USD 75 billion [48]. It is observed that losses amounted to USD 150 billion during the pandemic, and, subsequently, the MPC implemented an aggregate rate of 11.0% to combat the problem [51]. Thus, Pakistan’s economic development in the wake of this global epidemic necessitates more scrutiny.

2.4. Foreign Direct Investment (FDI) Impact

Since the peak of the 2007–2008 fiscal year, FDI inflows to Pakistan have been relatively low, decreasing from USD 4 billion in 2007 to USD 2.3 billion in 2018 [62]. Furthermore, FDI inflows increased by 68.3 percent to USD 1.34 billion in the first six months of the 2019–2020 fiscal year (July–December 2019), compared to USD 796.8 million the previous year, owing to the Chinese market’s dominance [45]. Nonetheless, according to [59,63], financial flows were higher near the end of the year as the Pakistani government abandoned its year-long

policy of allowing the rupee to weaken against the US dollar. The government's political vulnerability improved significantly because of the general election in July 2018.

However, the situation has changed due to the pandemic, as many governments have taken severe measures, and Pakistan is no exception [64,65]. Additionally, necessary financial arrangements have been made to ensure that the widespread health emergency can be paid for [66]. As a result, flexibility is critical during emergencies, as subsidiary and parent organizations require access to funds to operate [67,68]. Thus, FDI plays a significant role in supporting economies but might significantly be threatened by the pandemic's effects on FDI streams due to the worldwide lockdown.

2.5. Poverty, Unemployment, and Underemployment

According to the "Employment Trends" report published by the Pakistan Bureau of Statistics (PBS) in 2018, the total labor force in Pakistan stands at 63.4 million, of which vulnerable employment was 26.41 million (41.6%). Vulnerable employment is measured as the proportion of own-account workers and contributing family workers in total employment (poor workers generally dependent on daily wages). These workers are likely to be the largest impacted individuals and could lose their employment due to the COVID-19 pandemic [69]. The COVID-19 epidemic, according to [48], led in millions of job losses across the country. Because of the slowdown in the economic activity, which has placed employment in a precarious situation, there was a significant increase in poverty and unemployment [70]. According to a PIDE report, approximately 12.3 million people are expected to face unemployment under a scenario of moderate restrictions by the government [48]. According to [71], this is a proximately 46.3% of the total vulnerable employment and 19.4% of the total employment in Pakistan. Wholesale retail trade is expected to witness the highest layoff of 4.55 million people. Thus, the poverty rate in Pakistan could increase from 23.4% to 44.2%.

2.6. The Effect on the Travel and Tourism Industry

The tourism industry in pre-COVID-19 times was a worldwide, huge profit-earning industry [72]. As of late 2019, people's movements and the travel industry have been stopped by as much as 10.4% of the worldwide GDP, and 319 million people worldwide who work in the travel industry have lost their jobs as of late 2019 [73]. Remarkably, Pakistan is a varied geographical area, ranging from the coastline to the high mountain slopes of the Himalayas. Pakistan's travel industry began in 1972, with the creation of a range of services for different parts of the country [74,75]. However, in 2012, 46.7 million travelers went to Pakistan, and Khyber Pakhtunkhwa was 19%. During 2004–2011, the travel and tourism industry gradually improved and reached its peak in 2019 [69]. However, the tourism industry is now largely redundant, and many people will be unemployed because COVID-19 has natural and long-lasting ramifications for social interaction and entertainment.

2.7. The Effect on the Health Sector

Pakistan's border provinces were the first to experience the COVID-19 outbreak [66,76]. On 26 February 2020, Pakistan's Ministry of Health reported the first case of COVID-19 in Karachi, followed by Sindh province and Islamabad [77]. Later, a total of 98,943 confirmed positive cases with 4960 critical mortalities and 2002 deaths were reported by 7 June 2020. To ensure patients' good healthcare as much as possible, Punjab and KPK allocated more beds—955 and 856, respectively. However, [78] noted that Pakistan's current situation is unsatisfactory, requiring further action [79]. Other than China, the United States, the United Kingdom, and Russia, Pakistan is a developing country that lacks the financial resources to combat the COVID-19 pandemic. Furthermore, it does not have anywhere near enough hospitals and quarantine facilities, which are urgently needed [79]. Transmission of the virus can only be controlled by updating and disbursing the required medical facilities. For both arrivals and departures, Pakistan needs more screening facilities [77].

3. Methodology

The research methods mainly included a review of previous studies, determining Pakistan's economic recession because of the natural disaster; determining economic key indicators, such as GDP, imports and exports of Pakistan, interest rates, and remittances; and determining strategies for reducing poverty and unemployment because of the disaster. The study used a Keynesian approach to examine the impact of COVID-19 on public health and the economic consequences of this coronavirus outbreak [80,81]. According to the consensus, global supply chains are disrupted, and more factories will have to shut down because of the virus. In addition to affecting the country's overall production, this disruption has a direct impact on the country's aggregate demand. How can Pakistan's government deal with this situation through monetary and fiscal policy now that it has been raised? Keynesian and real-business-cycle models can be used to answer this question. COVID-19's supply disruption was the focus of this model, which we hope will be short-lived. A demand-driven downturn could result from the virus' spread, opening the door to stagnation traps set up by pessimistic animal spirits. [81].

4. Keynesian Theory and the 2020 Great Economic Recession

As a result of the novel coronavirus (COVID-19) spreading over the world, the World Health Organization (WHO) declared a worldwide pandemic. On the one hand, its effects on public health are likely to devastate all types of economic systems. This disruption affects aggregate demand and the overall level of manufacturing production and related industries. Now the question is: how can Pakistan's government cope with this situation through monetary policy and fiscal policy? To answer this, [42] suggests that it may be possible to solve economic breakdown for limited resources by implementing a modern financial strategy. This study investigated the real consequences of COVID-19 on the Pakistani economy using the Keynesian AD-AS theory, which is based on theoretical and logical premises [80–82]. According to Keynesian theory, COVID-19 has created a scenario in Pakistan where the supply and demand for goods and services have temporarily halted. The spread of COVID-19 has caused a reduction in demand (for goods and services), bringing countries on the verge of economic recession [81,82].

4.1. Coronavirus's Impact on Aggregate Demand (A.D.)

The new Keynesian model (stripped-down version) has been used as an economic theory standard [80]. As per the Keynesian theory, aggregate demand specifies employment and output levels. On the other hand, the concept of positive, productive growth determines the A.D. as faster productivity growth enhances future incomes and encourages the creators of goods and services to devote additional resources in the present to investment projects [83]. Aggregate demand consists of government purchases, investment, consumption, and net exports. Anything that changes C , I , $G.P.$, or $N.X.$ will shift aggregate demand.

The Keynesian theory says a recession is caused by an aggregate demand shock. At the end of February, the Pakistan government announced a complete lockdown to stop the virus's spread. After the announcement, all industries, hotels, gyms, offices, and borders were closed and restricted them from staying at home. This situation, firstly, hurts the aggregate demand at any level of prices. This situation made Pakistanis more pessimistic about their future survival because nobody knew when the case would become familiar. This situation decreased their current preferred consumption; consistently, they increased their current preferred saving.

On the other hand, the coronavirus (COVID-19) outbreak resulted in a long-term decrease in productivity. The government of Pakistan shut down all industries for a specific period. The other reason was that if the government permitted them to work, industrialists did not want them to work because of low demand. They could not afford the cost of production at low prices. Manufacturing and production investment are positively connected to collective needs. Businesses invest more when they expect to get high returns

on their investments, and the profit ratio is high when predictable demand is high. The decreases in investment further decrease production progress and, alternatively, collective need, and the cycle will continue. As shown in Figure 4, the coronavirus causes a temporary decline in production because of aggregate demand shocks.

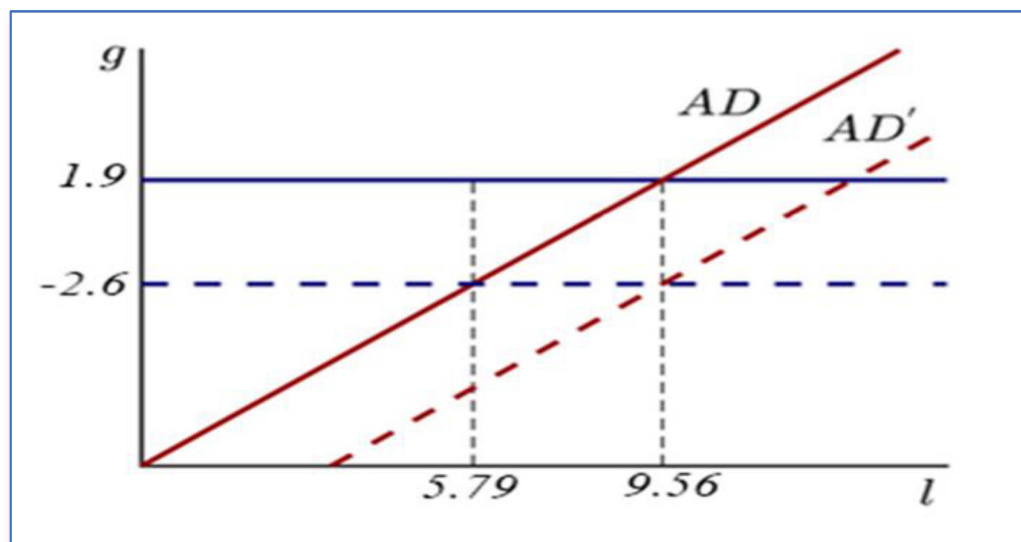


Figure 4. Impact of coronavirus on aggregate demand. Source: [83]. Note: unemployment on the horizontal axis; productivity growth rate on the vertical axis.

Already in the grip of a recession, the situation is deteriorating, particularly with the prospect of joblessness looming (up from 9.56 percent to 12.3 million to 18.53 million). These downsizings will likely reduce households' discretionary income while simultaneously causing a demand shock across the country. This risky condition diminishes not only local people's disposable income but also overall estimated remittances. A drop in private consumption expenditures of 4 to 8% is expected, affecting the country's economic picture and altering people's way of life as a result.

4.2. The Supply–Demand Doomsday Cycle

The beginning of COVID-19 badly impacted the domestic and international economy. This virus affected the demand side, and, at the same time, numerous companies were encountering various catastrophes with a specific ratio of income losses. Predominantly, businesses face multiple issues such as inventory-supply disturbances, termination of international trade orders, raw material scarcity, conveyance disturbances, and much more. Pakistan is a developing country, and small to medium industries play a vital role in boosting the trade sector. Furthermore, many small to medium businesses are fiscally vulnerable, with limited operations and resources. In comparison to their counterparts, i.e., huge corporations, their sensitivity to the current economic crisis is greater [84,85]. External financial situations, including epidemic diseases, earthquakes, floods, and other concerns, have unpleasantly impacted manufacturing activities and their existence. As we know, China is the world's largest export-oriented country, exporting raw materials and delicate, heavy machinery all over the world. According to the World Bank, almost 25% of world trade is affected by China's lockdown. Pakistan exchanged goods with China for approximately USD 11.8 billion in the 2019 fiscal year, but this was reduced to approximately USD 6 dollars in the current fiscal year during COVID-19 [86]. Pakistan's economic situation is very feeble after the four to five-month lockdown.

The Pakistan government announced a partial lockdown on May 31 to cope with this dire economic situation, but this strategy did not effectively overcome Pakistan's financial woes. Companies are functioning at less than half of their usual volumes because of the scarcity of inventory, a deficiency of labor, or limitations on people's travel forced

by the provincial governments, taking a toll on cumulative supply. In the lockdown, people's demand also reduces, and a decrease in demand reduces private investment and productivity [87]. This pandemic has become a supply-side shock in Pakistan primarily because the lockdown situation has directly affected the production of the goods-and-services sector, even though some highly contact-intensive sectors cannot fulfil individuals' private desires. Figure 5 depicts the absence of the Pakistan government, damaging monetary and economic policy participation. Such persistent stock disorder triggers a demand-driven decrease, triggering a supply and demand doom cycle and opening the door to sluggishness traps aided by unenthusiastic animal spirits.

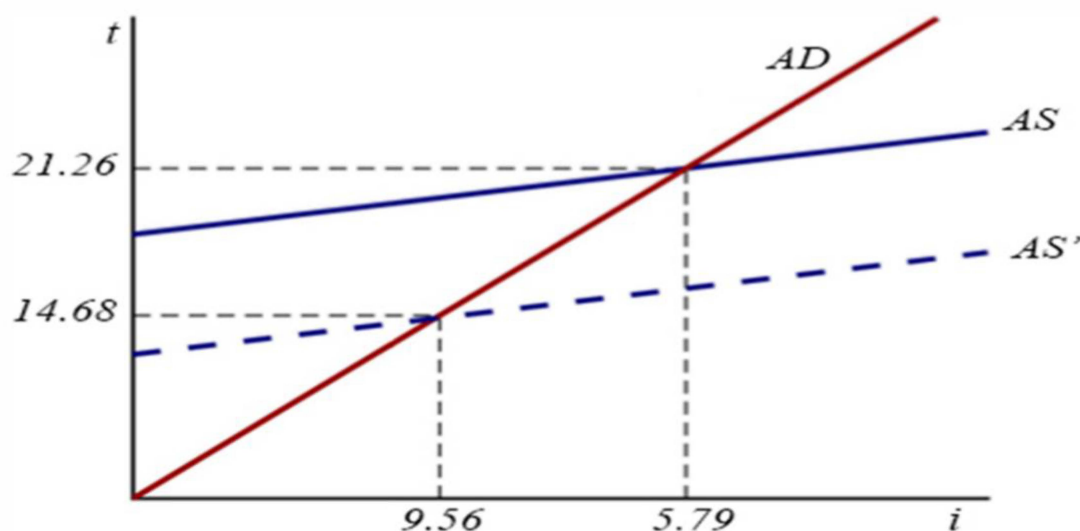


Figure 5. The supply–demand doom loop. Source: [83]. Note: unemployment on the horizontal axis; balance of goods trade on the vertical axis.

4.3. Animal Spirits and Stagnation Traps

In times of economic stress or uncertainty, the famous British economist John Maynard Keynes coined the term “animal spirits” to explain how people make financial decisions, such as purchasing and selling assets. In both the world and Pakistani economies, COVID-19 has left serious scars. The most prominent and most recent impact of the lockdown has been to create difficulties in current economic developments. Sindh was the first province to be placed under lockdown on 23 March 2020. In Pakistan, it is also a significant economic center. The country's most important industrial zone handled most international trade, accounting for more than 30% of total international trade. Due to fractional lockdown, just about 50 of Karachi's 2700 manufacturing units were functioning on the first working day [88]. The firms were facing many difficulties, like supply disruptions and the unavailability of workers. According to the recent survey conducted by the Pakistan Labor Force (2017–2018), the joblessness ratio in Pakistan was 5.8%, As for the ongoing current economic situation and lockdown, the joblessness ratio is likely to increase by 9.5% in the financial year 2020–2021 [89]. Small businesses, also known as self-employed individuals, include small shop holders, domestic companies, street sellers, and others, who severely depend on their daily business and are adversely impacted by the epidemic. According to the currently presented data, self-employed individuals total 35.7% (2017–2018) of the total Pakistan employment ratio [90].

Moreover, 59.6% (2019–2020) of these firms are defenseless. Furthermore, more than 87% of employment is in agriculture, a third–fourth of retail and wholesale business employees, more than 50% of restaurant employees, and others who work in business and real estate. The transportation and communication sectors employ more than two-fifths of the workforce [90]. Statistics on the sector-by-sector distribution of precarious employment in small industries are depicted in Figure 6 [91].

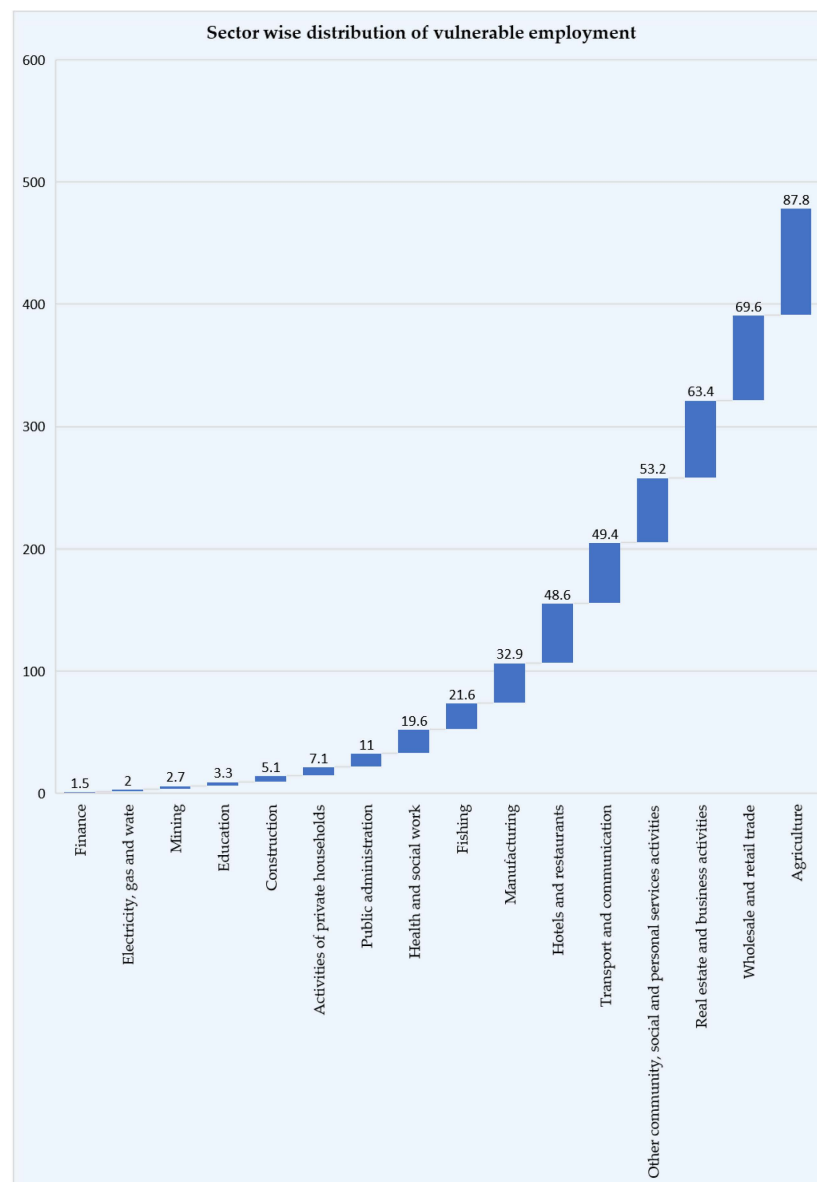


Figure 6. Distribution of vulnerable employment by sector. Source: [91].

On the other hand, many large- and medium-sized businesses are experiencing severe financial difficulties. The fabric and apparel industries, for example, have been severely harmed because of the unexpected lockdown. More than half of Pakistan's manufacturing sector's trade is in beverages, food, tobacco, and fabric; a drop in international demand for these industries will have a significant impact on Pakistan [61].

Similarly, the current crisis in other sectors is not diverse. Most prominently, in the blessed month of Ramadan, various industries such as fabrics, electronics, beauty, food, and shoes, along with others, were harshly impacted. These small- and mid-sized firms are likely to encounter liquidity crises and are severely hit by the current situation. However, those small businesses that are still operating their units face higher costs for buying masks to cover their faces, gloves, sanitizers, and other items to ensure their employees' health and shelter.

However, the above analysis indicates that Pakistan suffers disproportionately. So, the State Bank of Pakistan adopts a zero lower-bound-constraint monetary policy to mitigate the coronavirus's negative impact. The situation worsens because the money supply suddenly increases at a 7% interest rate, but the income level remains unchanged. In this

scenario, Pakistan's economy, which also faces a huge debt burden, subsequently faces a depreciation of the rupee, and upsurges the hyperinflation rate by 14.56 percent [71]. Then, Pakistan's economy will stick in a liquescency trap, and both progress and hire rates will be dejected auxiliaries.

Consider what might happen if Pakistani industrialists and consumers lose faith in future efficiency growth. Due to the zero lower bound, the SBP is unable to alleviate the corresponding reduction in demand. As a result, jobs and financial activity decrease, and firms respond by reducing savings, negatively impacting production growth. As shown in Figure 7, the coronavirus is widespread and can bring expectation-driven inaction traps, specifically by flagging the economy's evolution rudiments.

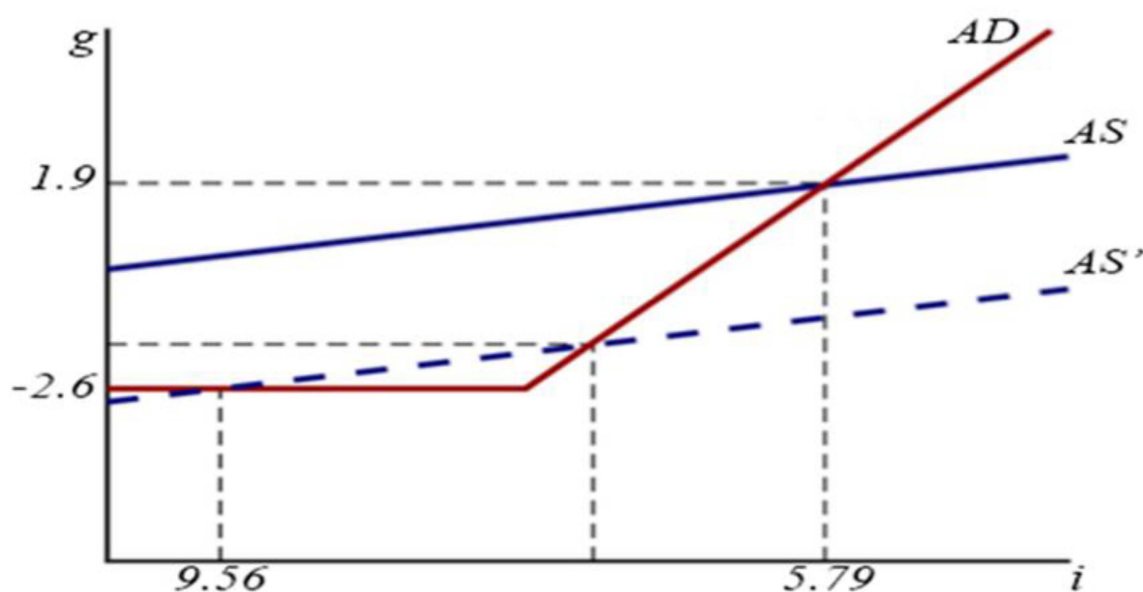


Figure 7. Animal spirits and stagnation traps. Source: [83]. Note: unemployment on the horizontal axis; productivity growth on the vertical axis.

An unaided conservative fiscal strategy may be insufficient or fail to converse in the loop when interest rates are near zero, as is debatably the case now. Dealing with such a situation will therefore require helicopter money or an expansionary monetary strategy, or both. When the economy is in a liquidity trap, helicopter money is a proposed unconventional monetary strategy that is occasionally recommended as an alternative to quantitative easing (QE) (when interest rates are near zero and the economy remains in recession). Although the term “helicopter money” was coined to describe central banks making direct payments to individuals, economists have used it to describe a variety of policy ideas, including the “permanent” monetization of budget deficits—with the added element of attempting to shock beliefs about future inflation or nominal GDP growth in order to change expectations.

4.4. Stabilization Policy under Keynesian Theory during Pakistan's Unsteady Economy

Following [85], the global economic uncertainty has triggered COVID-19, but the illustrative exercises and theories have future effects by aggregating business models. Hence, the method includes reviewing the previous studies of COVID-19 by following Keynes universal philosophy of occupation, interest, and coinage to determine the economic impact of the natural disaster on Pakistan, determining the vital economic indicators, such as GDP, imports, exports, interest rates, remittance, poverty, and unemployment and taking measures to reduce the economic impact by incorporating this set of factors into the development of concepts for dealing with and managing the epidemic. However, ref. [84]

makes evident that Keynes's theory significantly affects reassessing business models for overcoming economic crises.

4.4.1. Keynesian Policy Stabilization to Mitigate the Worse Effects of COVID-19

Now the query is: what strategy involvement can prevent the enchanting place's torpor trap? Because the policy rate is lowered due to the zero lower limit, a careful fiscal policy could only go so far. Still, under the burden of a huge debt, there is no room for Pakistan's state bank to reduce the interest rate to close to zero. It is the fifth time in four months that the State Bank of Pakistan has reduced the strategy rate (the per annum rate given in the Performance Swap Information Table) [86]. The State Bank of Pakistan (SBP) has now lowered the policy rate by a gigantic 625 points, from the comparatively extraordinary 13.25 pc to 7 pc, and this monetary action has shown its adverse effect. The price rise rate has hit a record of 14.56 pc, according to data published by Pakistan's official statistics department (PBS). According to Keynes (1958), only one financial strategy was ineffective in enhancing the economy—because decreasing interest rates governed it. In a depression, interest rates were previously adjacent to zero. Economic policy can be an influential support to transferring the economy because the outcome of an upsurge in outlays or a decrease in taxes would be multiplied by encouraging surplus demand for consumable goods by families.

4.4.2. Pakistan's Fiscal Policy in the Keynesian Legacy

The Keynesian model was established during the Great Depression as economic experts resisted elucidating the global financial failure and designing strategies to aid the economic recovery. This pandemic also created the situation of economic recession globally. The Pakistani government has made different fiscal changes to combat this financial crunch.

4.4.3. Increased Government Spending

Pakistan is amid a potential slump. Cash relocations and other economic measures may have an enormous optimistic effect on the international economy. Keeping in mind the current situation, the Pakistan government declared an aspiring financial inducement package in March intended to provide much-required aid to the helpless customers and firms within the realm. The USD 8 billion (PKR 1.2 trillion) aid package was designed to prevent Pakistan from falling into a self-perpetuating cycle of financial stress and reliance because of pre-emptive steps to curb the COVID-19 epidemic. Cash-transmission expenses under the Ehsaas Emergency Cash program, costing USD 894 million (PKR 144 billion), account for USD 894 million (PKR 144 billion) of the USD 8 billion economic incentive. Though this is a praiseworthy step, additional financial growth is still taking place [92,93].

The primary objective of this money injection is to unexpectedly increase household consumption, which immediately raises the demand for goods from the market and decreases the desired national saving at any level of real interest rate. As seen in Figure 8, the market curve moves from IS1 to IS2 to the right, but this capital infusion temporarily decreases the severity of the crisis. Other relief packages announced by the Pakistani government include PKR 572 billion for mass transportation, urban rail, and road transport systems; PKR 267 billion for solid waste management and storm water drainage systems; PKR 141 billion for sewage treatment plants; PKR 92 billion for water supply projects; and PKR 41 billion for disaster relief.

Like increasing government purchases, these fiscal stimuli eventually increase market demand and allow industrialists to invest. It has also decreased gas charges for exports concerned with zero-rated activities in July and August to help them recommence their construction activities after the COVID-19-associated lockdown [94]. So, this fiscal-policy change adjusts prices because companies encounter the developed demand at the static price level and shift the market demand curve upward. This expansionary change raises output and employment.

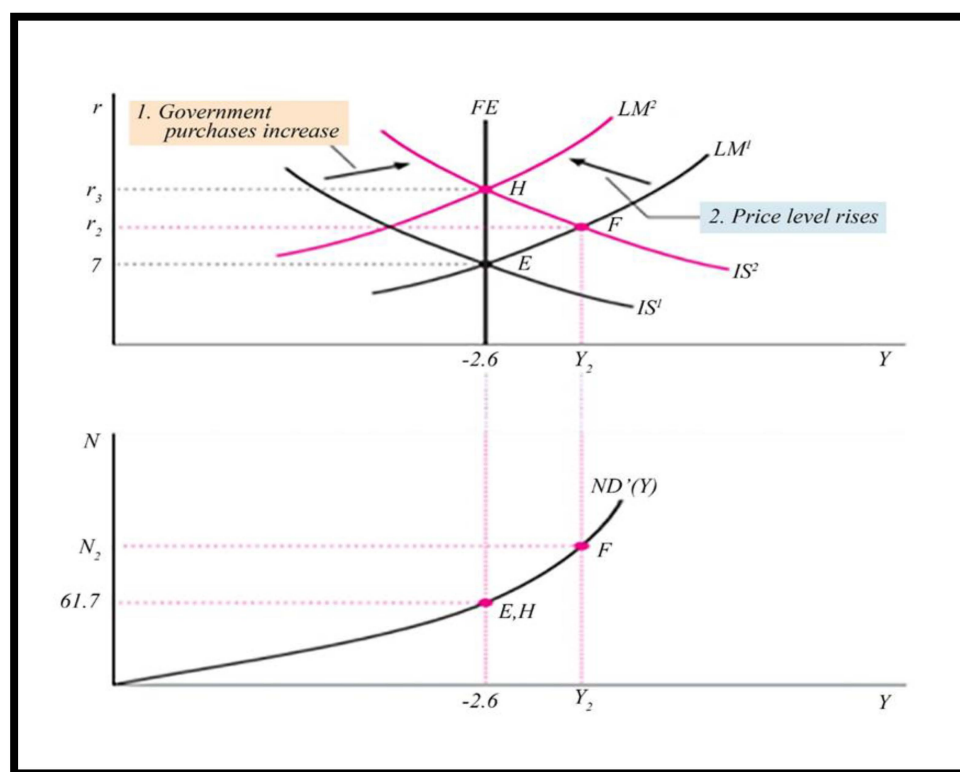


Figure 8. An increase in government purchases. Source: [83]. Note: interest rate and employment on the vertical axis; output on the horizontal axis.

5. Development of a Proposed Study Framework

The proposed research framework suggests focusing on the economic policies that have prolonged the medical crisis, as well as a grouping of financial, economic, and other systems and border cessations, self-isolation, and social connection constraints to resolve global and monetary market disputes. As a result, the study draws a framework in Figure 9 that is portrayed graphically with the discussions.

Policy Challenges

The problems for economic experts have been one of the uses of besieged policies that can address what are likely to be short-term anomalies in countries that can survive the impact of the epidemic itself without forming falsifications. Experts around the world have been left speechless by the quick shift in the condition of the global health emergency, which appears to be evolving into an international corporate trade and economic disaster with increasing consequences for the global economy. As the epidemic's fiscal impact grows, experts are focusing increasingly on remedies that, rather than longer-term considerations like debt accretion, may address immediate monetary consequences. Because of the low elasticity of fiscal and monetary support under conservative standards, many strategists have restricted their capacity to respond to the current crisis, given the corresponding recent decline in global economic development, particularly in output and trade, since the beginning of the flu epidemic. The epidemic has an influence on international policymaking. In the beginning, the fiscal impact of the pandemic was likely to be a temporary supply of products as factory production stopped because workers were self-quarantined by social contact to decrease the scope of the outbreak.

These rising financial impacts are likely to increase liquidity constraints and credit-market constraints in international monetary markets as businesses accumulate cash, with adverse effects on economic growth. Unlike the financial crisis of 2008–2009, lower consumer demand, labor market issues, and a decline in company operations rather than a dicey industry by foreign banks have contributed to commercial credit risks and potential

bankruptcy. These undercurrents in the economy have led some viewers to inquire if these incidents are the beginning of full-scale international monetary issues.

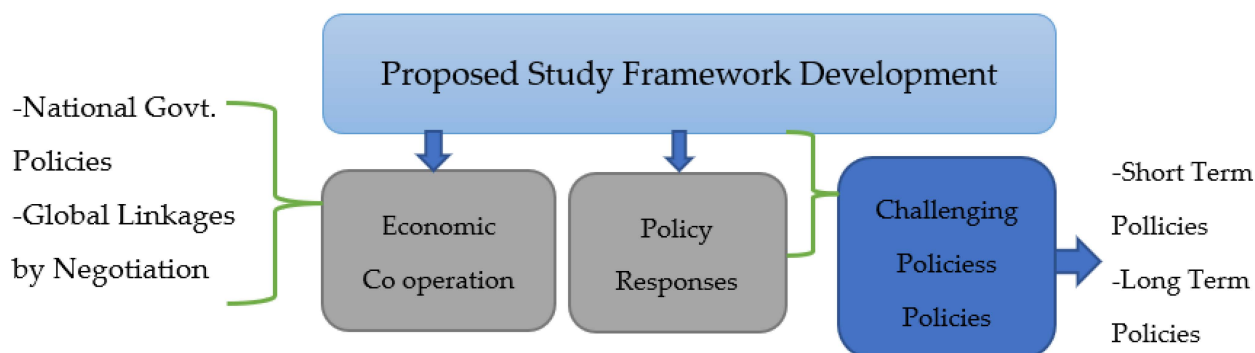


Figure 9. The proposed study framework for major development. Source: author's development (2020).

Early in 2020, economic growth in the world's most-developed and fastest-growing economies was modest but encouraging. Nations exceedingly reliant on trade, such as Germany, Japan, Mexico, Canada, Italy, and South Korea, and exporters, are now reported to be most adversely affected by the sudden decrease in economic activity because of COVID-19. The OECD discovered that China's manufacturing cuts had a global impact on circulation [62]. Because of strong recoveries in a few key economies, the global economy is expected to grow at a rate of 5.6 percent in 2021, the fastest rate of recovery in 80 years. According to the World Bank's June 2021 Global Economic Prospects, many emerging markets and developing nations are still grappling with the COVID-19 pandemic and its consequences. Despite the rebound, worldwide output will be around 2% lower by the end of the year than it was before the pandemic. For nearly two-thirds of emerging markets and developing nations, per capita income losses will not be recovered by 2022. The pandemic effects have reversed poverty-reduction gains and exacerbated insecurity and other long-standing concerns in low-income economies where vaccination has lagged.

According to a new survey, Asia's economy could be in for a slowdown by 2020. In addition, the development ratio in Latin America and the Middle East is expected to be negative in 2020 [62]. These predictions are likely to be revised downward soon because of COVID-19's influence on international commerce, low energy and product costs, and a rise in the dollar's value.

Another result will be a slow rate of economic growth and a reduction in development. Due to a drop in trade with China and low commodity prices, goods-exporting countries are likely to see a greater decline in development than expected in previous estimates. Simultaneously, many countries are artificially enhancing their expenditure to battle the epidemic and will encounter a sudden decline in their income level, stressing public finances and increasing the chances of sovereign (government) evasion.

When foreign currency is used to indicate debt obligations, deficit crescendos are a particular concern in developing countries (such as U.S. dollars). The value of most developing economies' currencies has plunged since the onset of the epidemic, raising the debt's value in terms of native currency.

Most economies have shown a pledge to or have already instigated strategies to uplift economies affected by the epidemic. Several decisions were made by these countries on the sort of assistance to offer (loans versus direct payments), the amount of assistance required, how to define resources, and the terms to be accorded to funding, if any. To combat the crisis, several unanticipated fiscal and monetary measures have been enacted in countries around the world. These economies are constrained by a lack of financial resources, and the medical care system has become overburdened and unprotected in many cases.

As the COVID-19 issue is global and all countries are affected to a great degree by the downturn or forthcoming recession, global leaders need to overcome the COVID-19 effects swiftly and forcefully on their respective economies and their economic impact. Thus, global action is required that will lay down a proposed framework with international cooperative macroeconomic policies to restore confidence and to address the feasible solution to the existing barriers of the post-COVID crisis to overcome the great recession expected by using collective action with the following research questions:

- How do national governments negotiate and collaborate to overcome sharp drops in income, putting a strain on their public finances, and the looming possibility of sovereign (government) evasions and a 2021–2025 recession?
- How can Pakistan and Canada leverage global cooperation to avoid a sudden drop in income, a strain on their public finances, and an increase in the likelihood of sovereign (government) evasions?

6. The Future Study Recommendation

6.1. Significant Economic Development

The current critical situation demands greater attention paid to the industry, and Pakistan's economic sector consists of small- to medium-sized enterprises (SMEs). The world economy was one in which SMEs received loans to cover their inactivity periods for three months, while special funds were allocated to the larger firms with short-term loans [95]. However, SMEs in Pakistan are facing severe financial difficulties, which are jeopardizing their business prospects [96]. Other governments, such as France and Germany, plan to address the panic-like situation caused by COVID-19 with a liquidity program based on short-term refinancing schemes, accelerated payments, and tax credits [43,44,46]. Similarly, the Government of Pakistan is considering offering tax relief and other packages to compensate for the significant losses inflicted at present [42].

Therefore, Pakistan may follow other countries' strategies to save the economy and avoid the appalling social fallout. Interestingly, China's economic system is to stabilize its economy, and, as one example, Alibaba and J.D. Com's talent-sharing program in China has assisted in reducing the effect of the epidemic. They have recruited short-term workers to temporarily affect businesses. Furthermore, the European Tourism Manifesto Alliance [94,96] advocated for national governments to provide temporary assistance to the tourism and travel industry in the form of short- and medium-term loans. This alliance has proposed lowering or waiving travelers' taxes [97,98]. Several companies in the U.K. now provide online services and virtual sessions to education bodies so they can continue their studies without interruption [99]. The Government of Pakistan and SBP are partnering to provide relief to the industry and the wider community [51].

6.2. Policy Implications for Selected Sectors

At the start of the crisis, many economists felt constrained by the need to respond to the catastrophe. The global synchronized slowdown in economic growth, especially manufacturing and trade, that had preceded the viral breakout, was achieved at this time due to the partial elasticity of the currency and fiscal sustenance under predictable criteria. Liquidity and credit-market problems have also prompted financial analysts to deal with supply-side restraints with a particular range of experience.

Before banking institutions give credit deferment or a financial aid system is in place, job losses may result in missed mortgage and rent payments. Consequently, mortgage evasion may adversely disrupt the mortgage-backed-securities sector and the accessibility of mortgage funds, and it may have a disruptive effect on the country's economic growth rate. Globally, losses in the valuation of many financial markets impact the wealth of households, especially that of fixed-income retirees and those who own securities. Investors who operate on mortgage-backed assets have reduced their holdings. In the current environment, rate-cutting markets have not always routinely handled classic policy instruments

like monetary assistance. Such volatility raises the question of what nations should do to fix the international economy's weaknesses.

Although the Pakistani government has done its best, it should still take some measures to tackle this pandemic. Firstly, in this emergency, supportive macroeconomic policies are required to restore trust and economic recovery. Enterprise tax credits should be given so that employers can pay workers their wages and earn an income themselves. Interest-free loans to companies may help address this situation, at least in the short term. We also must provide masks, disinfectants, and medical equipment to manufacturers, even though there are several downsides to doing so. Thirdly, online shopping will grow further and be encouraged to retain social distance and prevent people from risking the spread of COVID-19 when they leave their homes. Finally, a decline in the trade flow of textiles to China, the U.S., and the E.U. could be of great advantage to Pakistan. It can now dictate the price, availability, and routes of low-cost articles according to market conditions. In a nutshell, from the Keynesian theory study, Pakistan already falls into unfavorable circumstances that would lead to deterioration.

Despite the Pakistani government's announcement of compensation packages for catastrophe victims, more substantial measures are still required to get the economy back on track once the crisis has passed. However, the suggested necessary steps are given below:

1. In this emergency, supportive macroeconomic policies are required to restore trust and demand recovery. Hence, the enterprise tax credit may be provided to employers for collecting wages. Interest-free loans to companies may help address the lost income situation.
2. The healthcare industry has made a significant amount of money by mass-producing masks, hand sanitizers, and surgical supplies in large quantities, despite its many shortcomings. The other industry facing a recession could step forward to produce medical equipment. Furthermore, it does not have anywhere near enough hospitals and quarantine facilities, which are urgently needed [79,100,101]. Transmission of the virus can only be controlled by updating and distributing the required medical facilities. For both arrivals and departures, Pakistan needs more screening facilities [77,102].
3. Online shopping can be in vogue and will grow further, keeping in mind the prevailing unpredictable situation. In addition, the tourism industry is now largely redundant, and many people will be unemployed because COVID-19 has natural and long-lasting ramifications for social interaction and entertainment.
4. A decline in the textile sector's trade flow among China, the U.S., and the E.U. can be of great advantage to Pakistan, as Pakistan can manipulate some of the low-cost articles according to her choice.

7. Conclusions

Countries throughout the world are undergoing economic trauma caused by a global pandemic. Millions of people working in various industries are waiting impatiently for COVID-19 to end and never emerge again. Simultaneously, people are trying their level best to cope with this situation, mainly affecting the wage earners and those running small business enterprises. Still, it cannot do so due to curfews and lockdowns. Many industries have fallen prey to this scourge, and it means that GDP, remittances, interest rates, exports, imports, tourism, and health cannot function properly, if at all. The Islamic Republic of Pakistan is no exception, as it is now greatly distressed and experiencing a grave economic crisis due to COVID-19. Pakistan is currently under negative circumstances, according to Keynesian theory, and to avert further deterioration, the government has established special assistance packages for the victims of this disaster. More drastic measures will be required to get the economy back on track once the crisis has passed. Nonetheless, the fiscal policy can be a powerful device to alter the economic condition because the increasing trend in spending or a diminishing tendency in the taxes would be multiplied by encouraging supplementary demand for families' depletion of goods.

This means that reducing COVID-19 transmission while also having a positive impact on the environment, sustainable production, society, the economy, and worldwide prevention measures can be the key objectives of future work. Thus, for the study purpose, the methodology utilized in this exploratory study started with a review of how the COVID-19 pandemics exacerbated global health, identifying the changeable features and structural alterations that made a difference in the built environment, and then analyzed examples of such infections and changes to apply global new rules and regulations, identifying what controllable strategies and anticipations emerge from rethinking sustainable production.

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